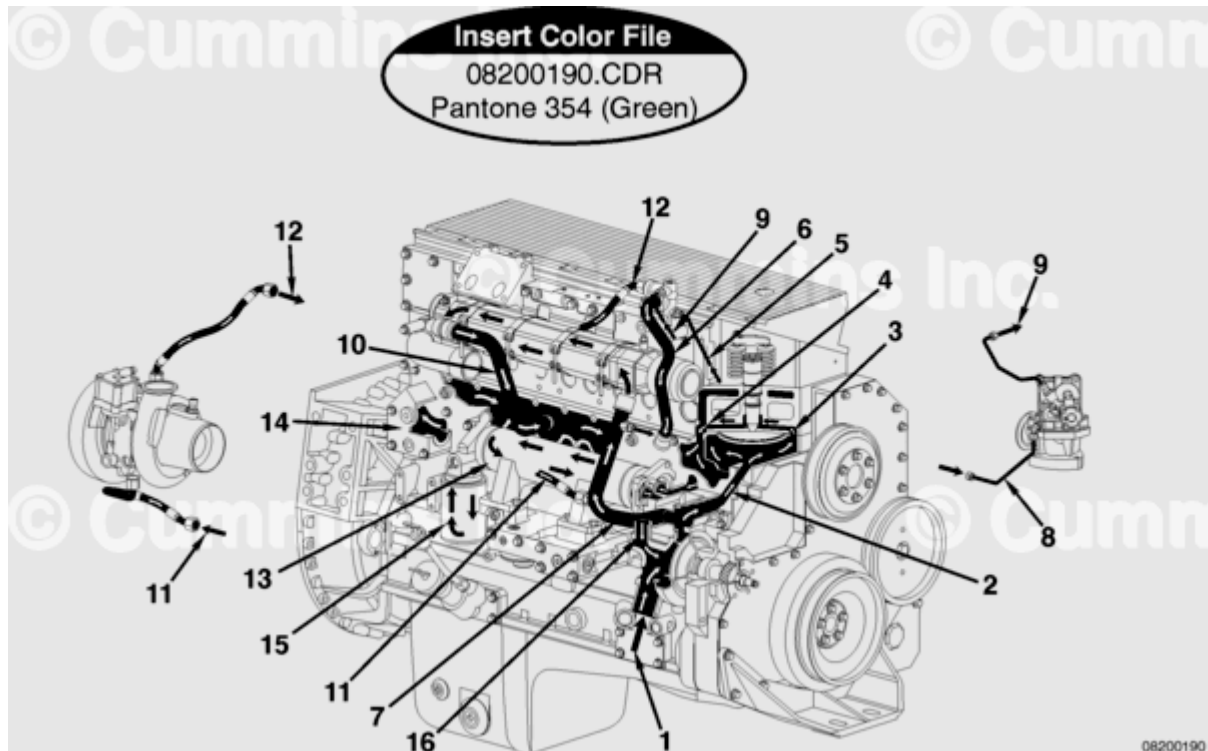


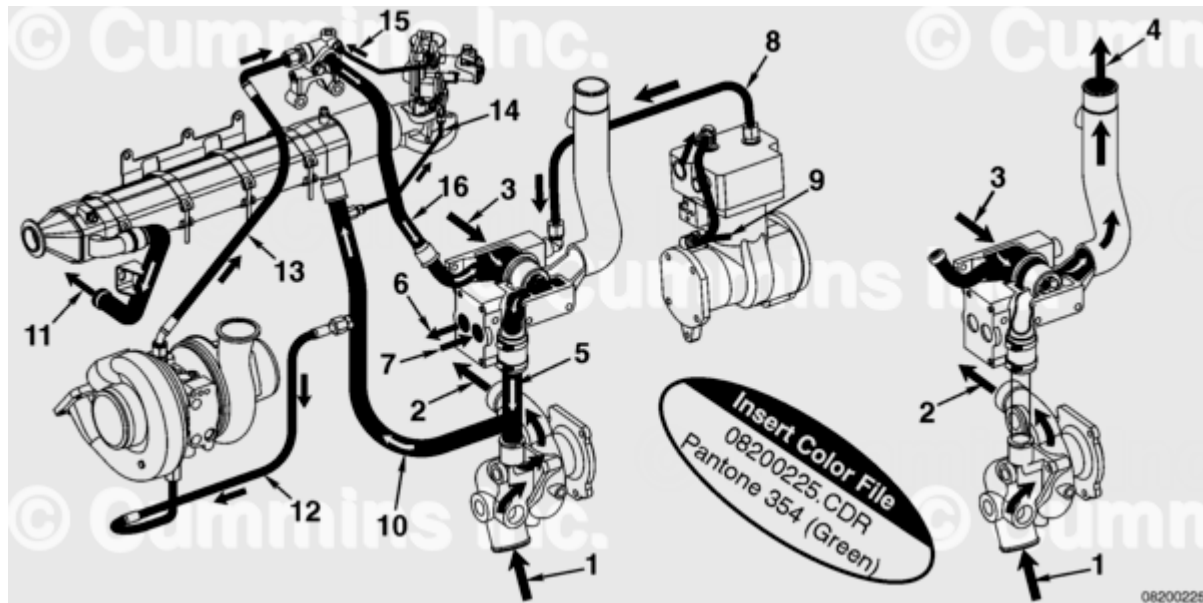
Trainings ▾

Flow Diagram

With CM870 and CM570



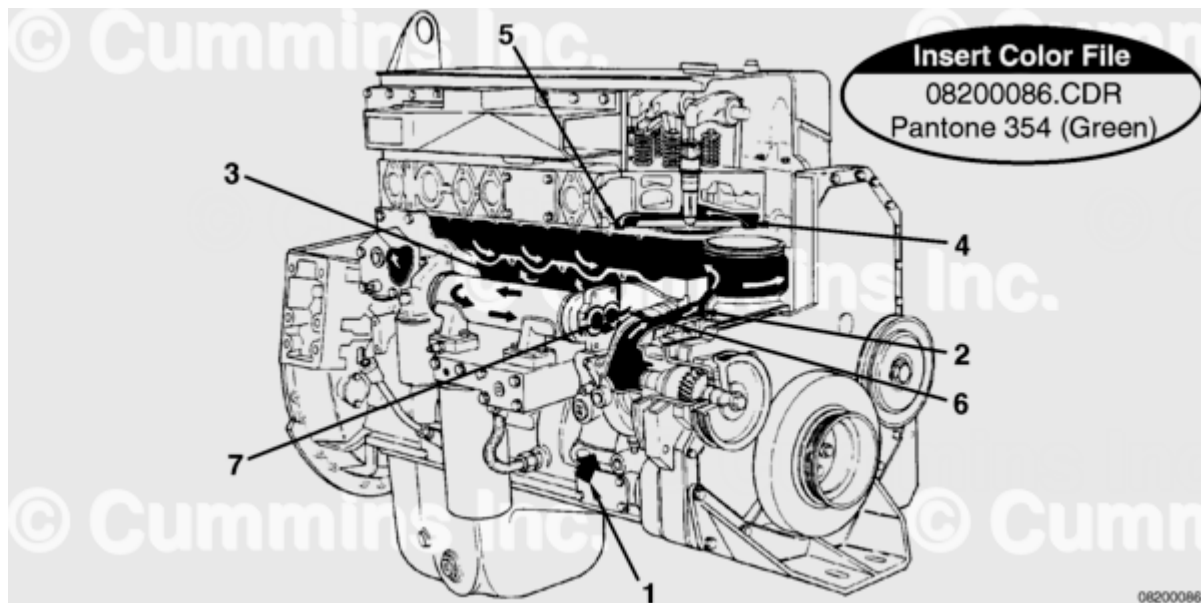
1. Water pump coolant inlet
2. Coolant supply to block
3. Coolant supply to cylinder head
4. Coolant return to low pressure cavity
5. Cylinder head coolant vent to water manifold
6. Coolant return to thermostat housing
7. Coolant supply to EGR cooler
8. Coolant supply to EGR valve
9. EGR valve coolant return to water manifold
10. EGR cooler coolant return to low pressure cavity
11. Coolant supply to turbocharger
12. Turbocharger coolant return to water manifold
13. Oil cooler
14. Heater housing
15. Coolant filter, if equipped
16. Coolant return from thermostat housing.



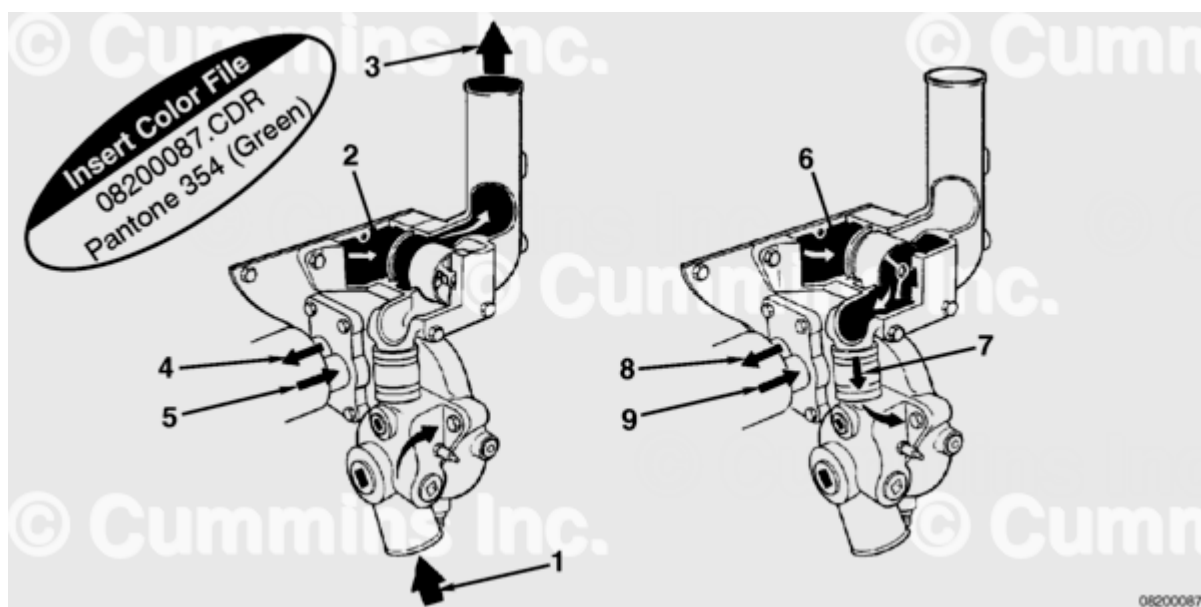
Thermostat Open and Closed

1. Water pump coolant inlet
2. Coolant to block
3. Upper manifold cavity (coolant to thermostat)
4. Coolant outlet (coolant to radiator)
5. Coolant bypass (return to water pump)
6. Lower manifold cavity to lubricating oil cooler
7. Lubricating oil cooler to upper manifold cavity (before thermostat)
8. Coolant from air compressor
9. Coolant to air compressor
10. Coolant supply to EGR cooler
11. Coolant from EGR cooler to block
12. Coolant supply to turbocharger
13. Coolant from turbocharger to water manifold
14. Coolant supply to EGR valve
15. Coolant return from EGR valve to water manifold
16. Coolant return from water manifold to thermostat housing (before thermostat).

Automotive Applications Without EGR



1. Water pump coolant inlet
2. Coolant to lower manifold cavity
3. Coolant to cylinder liner block cavity
4. Coolant to cylinder head
5. Coolant to upper manifold cavity
6. Lower manifold coolant to oil cooler
7. Oil cooler to upper manifold cavity.



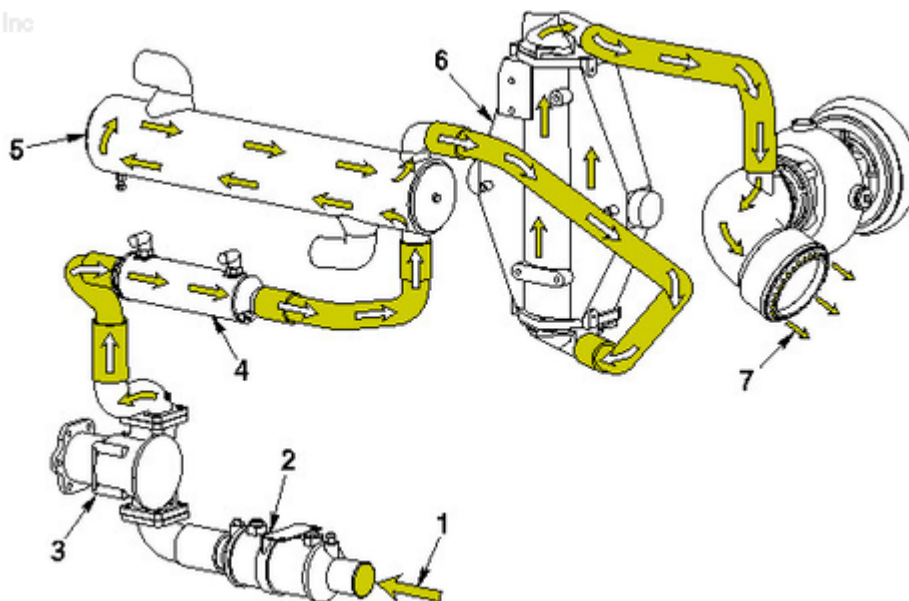
Thermostat Open and Closed

1. Water pump coolant inlet
2. Upper manifold cavity (coolant to thermostat)
3. Coolant outlet
4. Lower manifold cavity to oil cooler
5. Lubricating oil cooler to upper manifold cavity (before thermostat)
6. Upper manifold cavity (coolant to thermostat)

7. Coolant bypass (return to water pump)
8. Lower manifold cavity to oil cooler
9. Lubricating oil cooler to upper manifold cavity (before thermostat).

Marine Applications

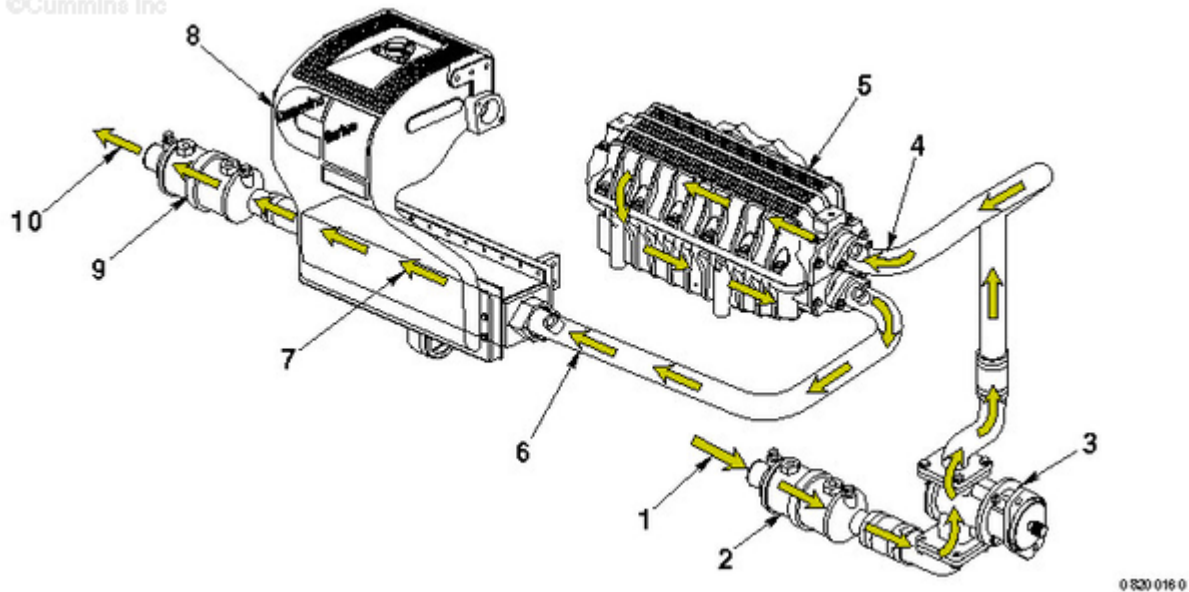
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Sea Water Flow (Heat Exchanger)

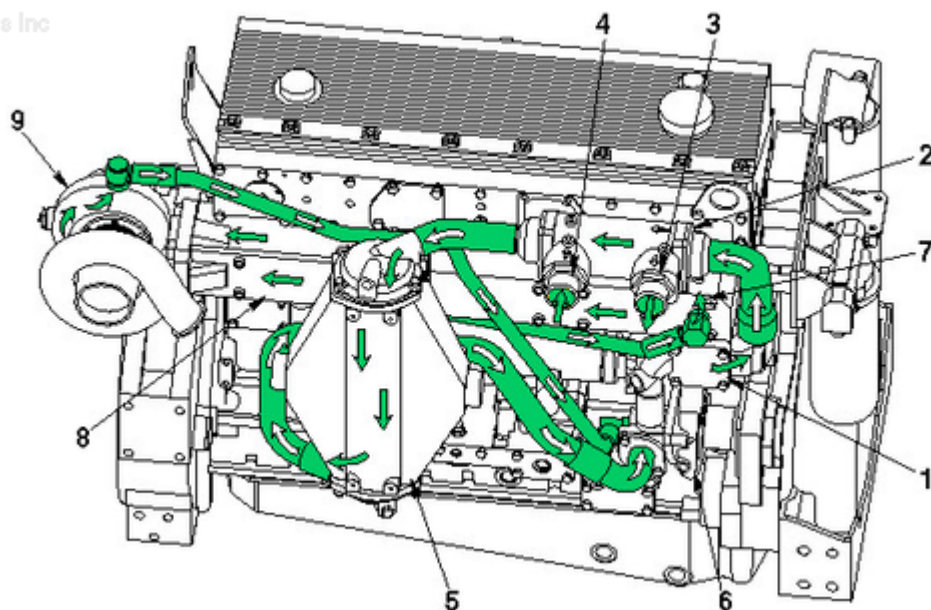
1. Sea water inlet
2. Fuel cooler
3. Sea water pump
4. Gear oil cooler
5. Heat exchanger
6. Aftercooler
7. Sea water outlet.



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Sea Water Flow (Heat Exchanger)

1. Sea water inlet
2. Fuel cooler
3. Sea water pump
4. Sea water pump to aftercooler
5. Aftercooler
6. Aftercooler to heat exchanger
7. Heat exchanger
8. Expansion tank
9. Gear oil cooler
10. Sea water outlet.



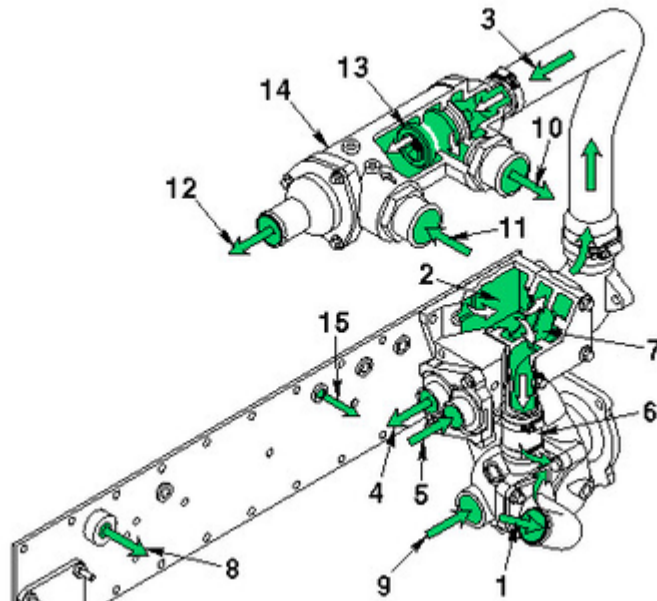
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Coolant Flow, Keel Cooled (one of three)

1. Engine thermostat cavity (thermostat removed)
2. Keel cooled thermostat housing

3. Coolant out to keel cooler
4. Coolant in from keel cooler
5. Aftercooler
6. Water pump
7. Coolant flow to exhaust manifold
8. Water cooled exhaust manifold
9. Water cooled turbine housing.

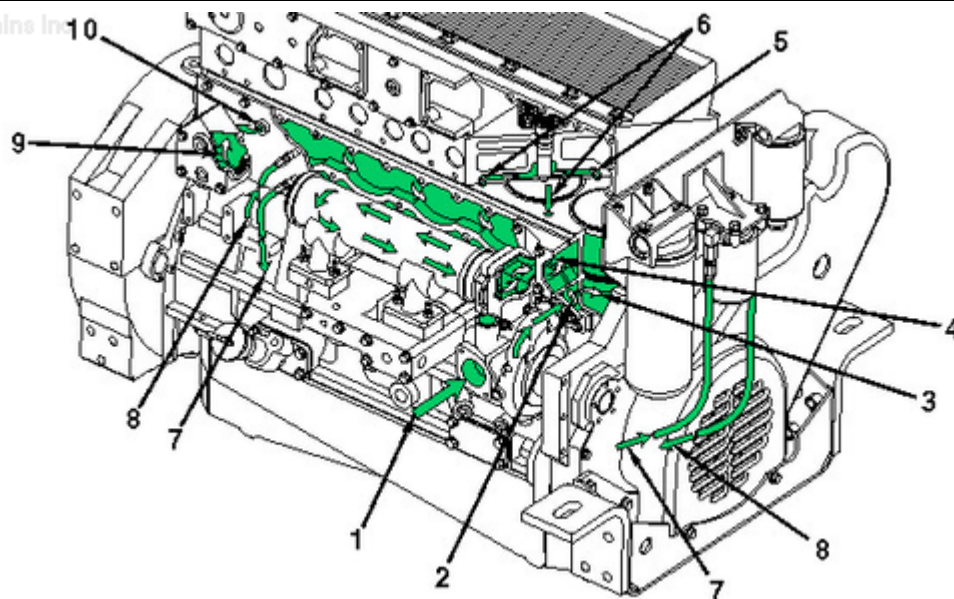
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Coolant Flow, Keel Cooled (two of three)

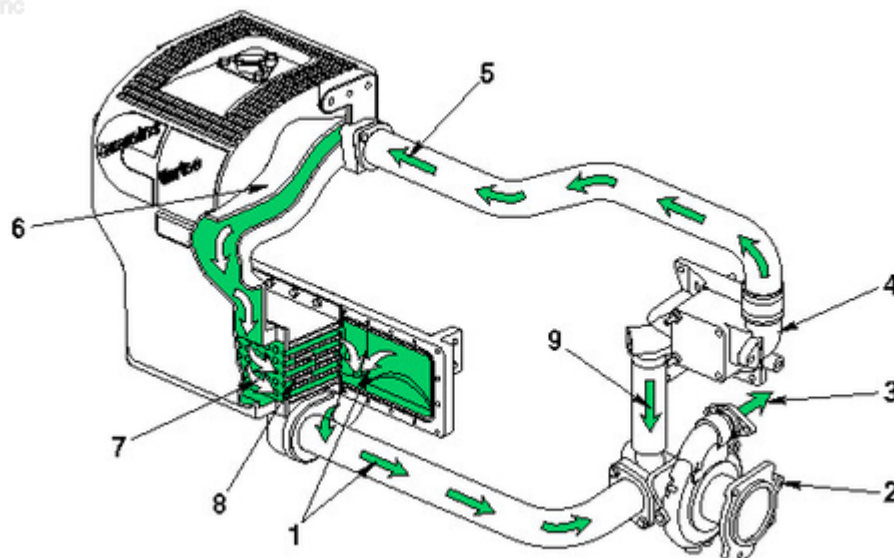
1. Water pump coolant inlet
2. Upper manifold cavity
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat)
6. Coolant bypass hose - plugged
7. Engine thermostat cavity (thermostat removed)
8. To exhaust manifold
9. Turbocharger coolant return to water pump inlet
10. To keel cooler
11. From keel cooler
12. To aftercooler
13. LTA thermostat
14. LTA thermostat housing
15. To turbocharger.



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Coolant Flow (three of three)

1. Flow to water pump
2. Flow to oil cooler and lower manifold cavity
3. Flow from lower manifold cavity to cylinder liners
4. Flow from oil cooler to thermostat housing
5. Flow to cylinder head
6. Flow from cylinder head to upper manifold cavity
7. Flow to coolant filter
8. Flow from coolant filter
9. Coolant heater - optional
10. Upper manifold to exhaust manifold.

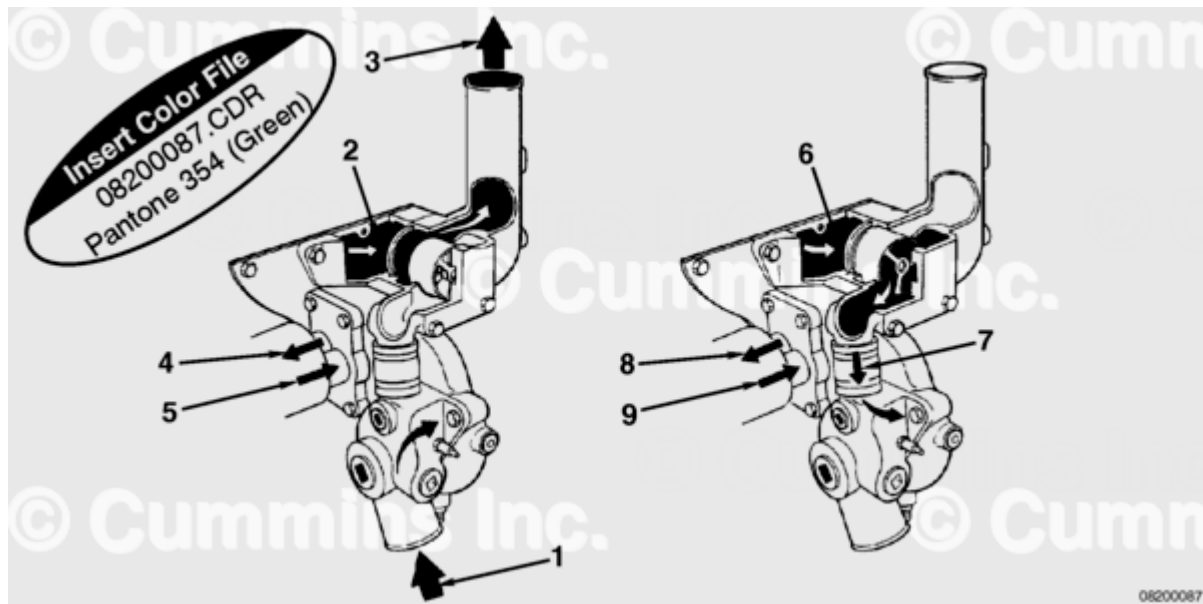


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Coolant Flow, Heat Exchanger (one of three)

1. Coolant flow to water pump
2. Water pump

3. Coolant flow to lower manifold cavity
4. Thermostat housing
5. Thermostat housing to heat exchanger
6. Expansion tank
7. Heat exchanger sea water core
8. Coolant flow through sea water core
9. Coolant bypass.

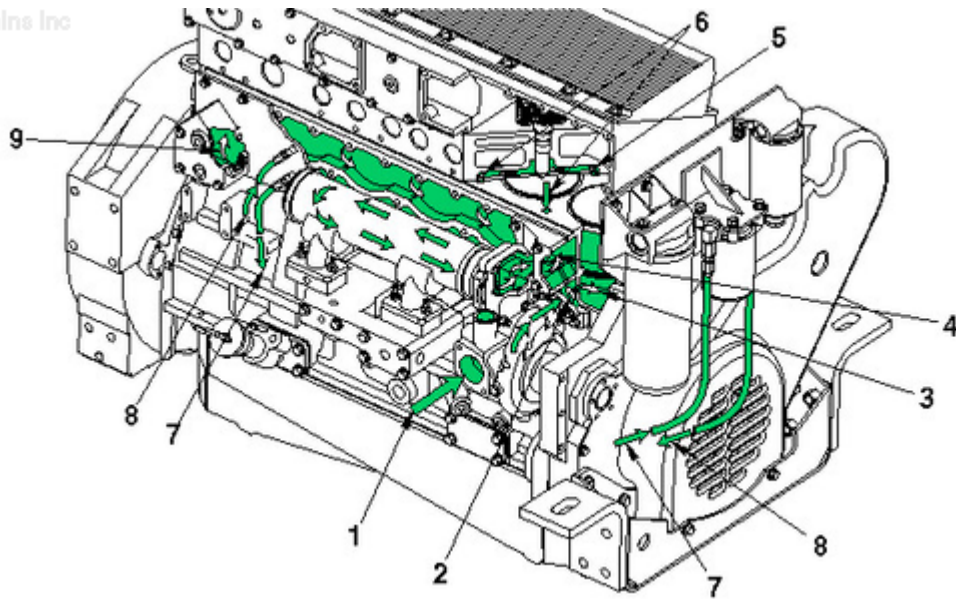


Coolant Flow, Heat Exchanger (two of three)

Thermostat Open (1-5)

Thermostat Closed (6-9)

1. Water pump coolant inlet
2. Upper manifold cavity
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat).
6. Upper manifold cavity (coolant to thermostat)
7. Coolant bypass (return to water pump)
8. Lower manifold cavity to cooler
9. Cooler to upper manifold cavity (before thermostat).

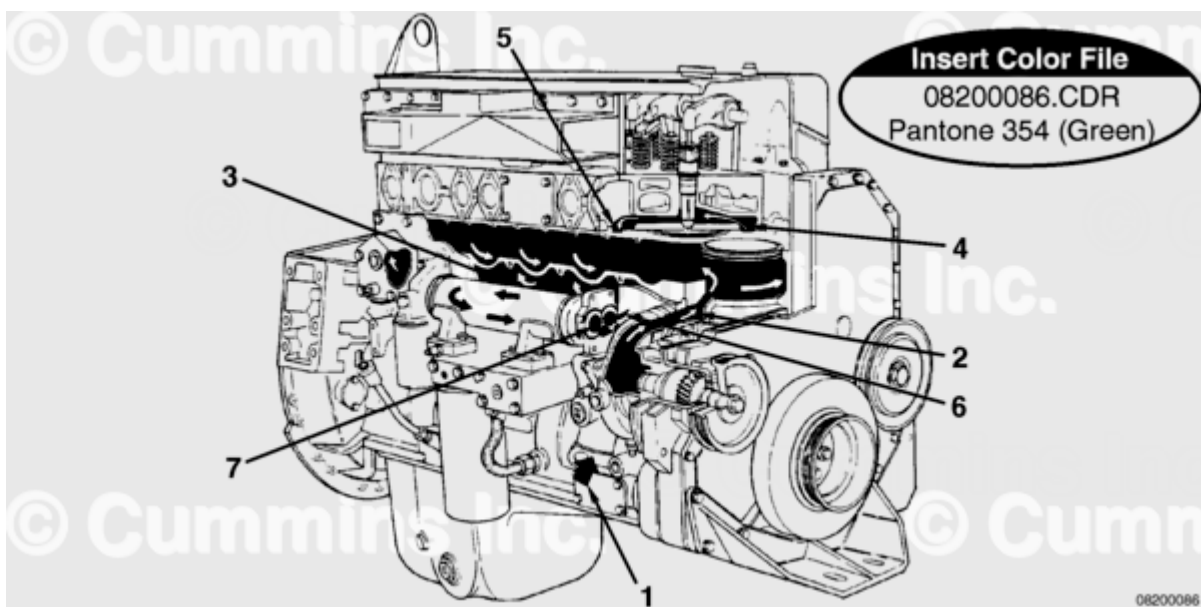


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Coolant Flow, Heat Exchanger (three of three)

1. Flow from heat exchanger to water pump
2. Flow to oil cooler and lower manifold cavity
3. Flow from lower manifold cavity to cylinder liners
4. Flow from oil cooler to thermostat housing
5. Flow to cylinder head
6. Flow from cylinder head to upper manifold cavity
7. Flow to coolant filter
8. Flow from coolant filter
9. Coolant heater - optional.

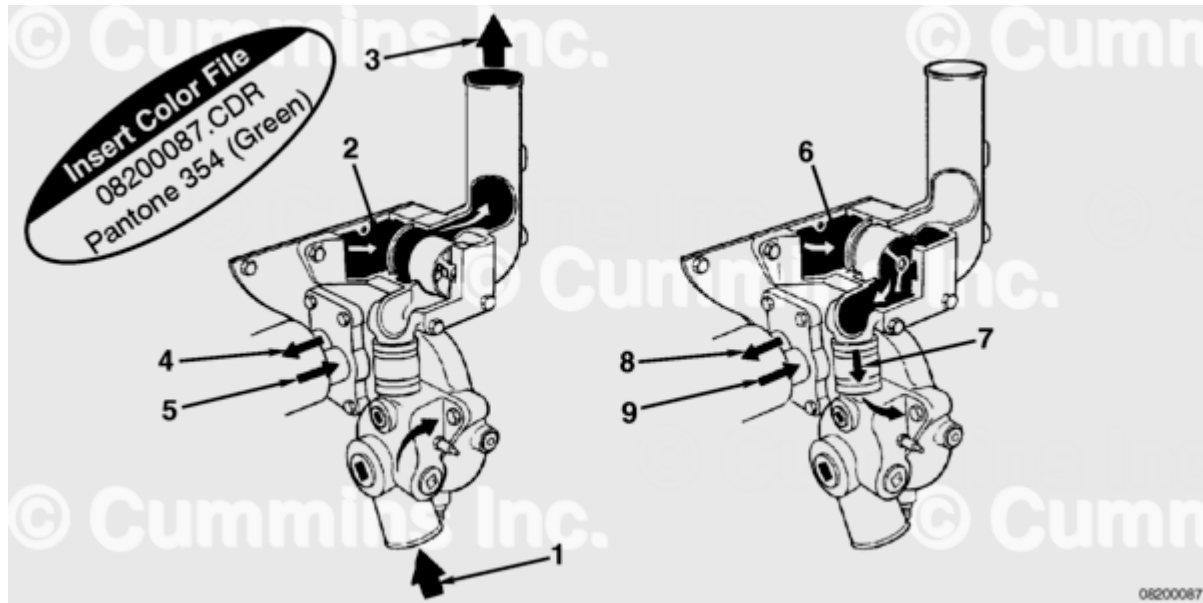
Industrial and Generator Drive



08200086

1. Water pump coolant inlet
2. Coolant to lower manifold cavity
3. Coolant to cylinder liner block cavity

4. Coolant to cylinder head
5. Coolant to upper manifold cavity
6. Lower manifold coolant to oil cooler
7. Oil cooler to upper manifold cavity.



Open (1-5)

Closed (6-9)

1. Water pump coolant inlet
2. Upper manifold cavity (coolant to thermostat)
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat).
6. Upper manifold cavity (coolant to thermostat)
7. Coolant bypass (return to water pump)
8. Lower manifold cavity to cooler
9. Cooler to upper manifold cavity (before thermostat).